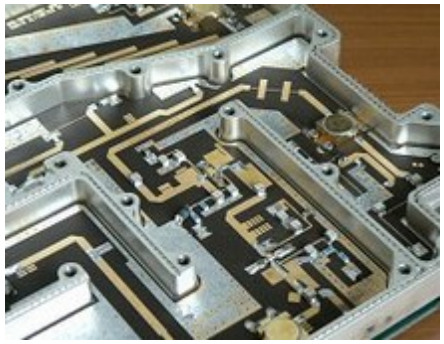
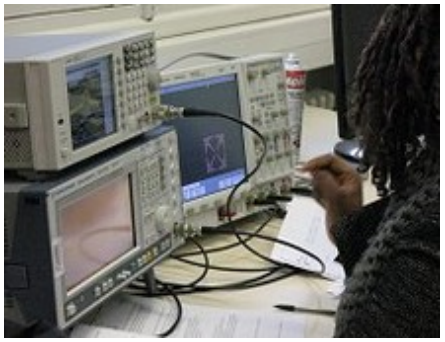
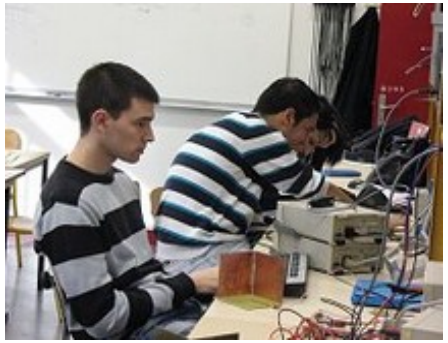
Notions sur les radars
Aspect pratique







Impulsion Radar
Domaine temps, domaine fréquence

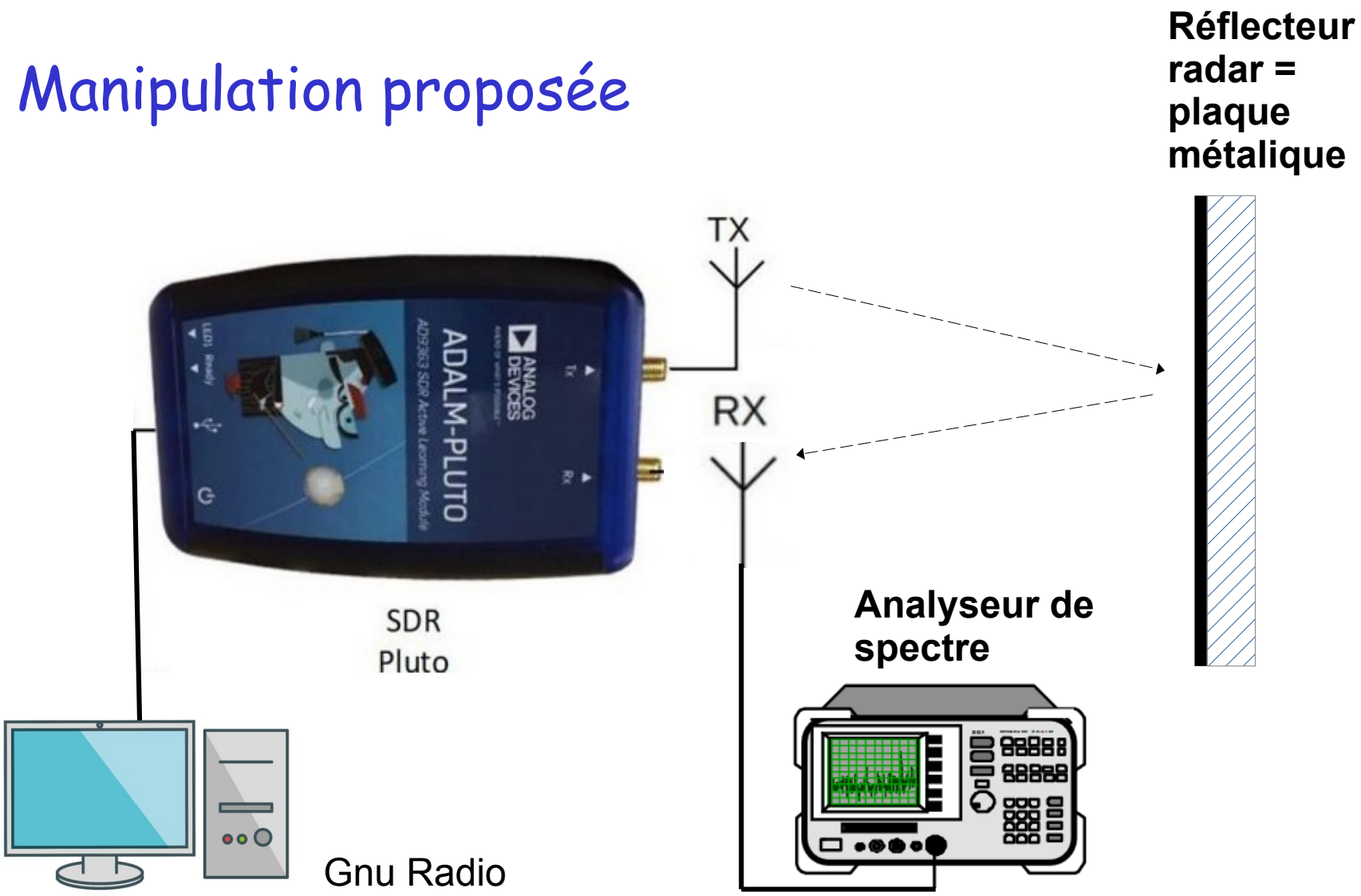
Franck Daout
fdaout@parisnanterre.fr

https://cva-geii.parisnanterre.fr/

CFD - Bourges

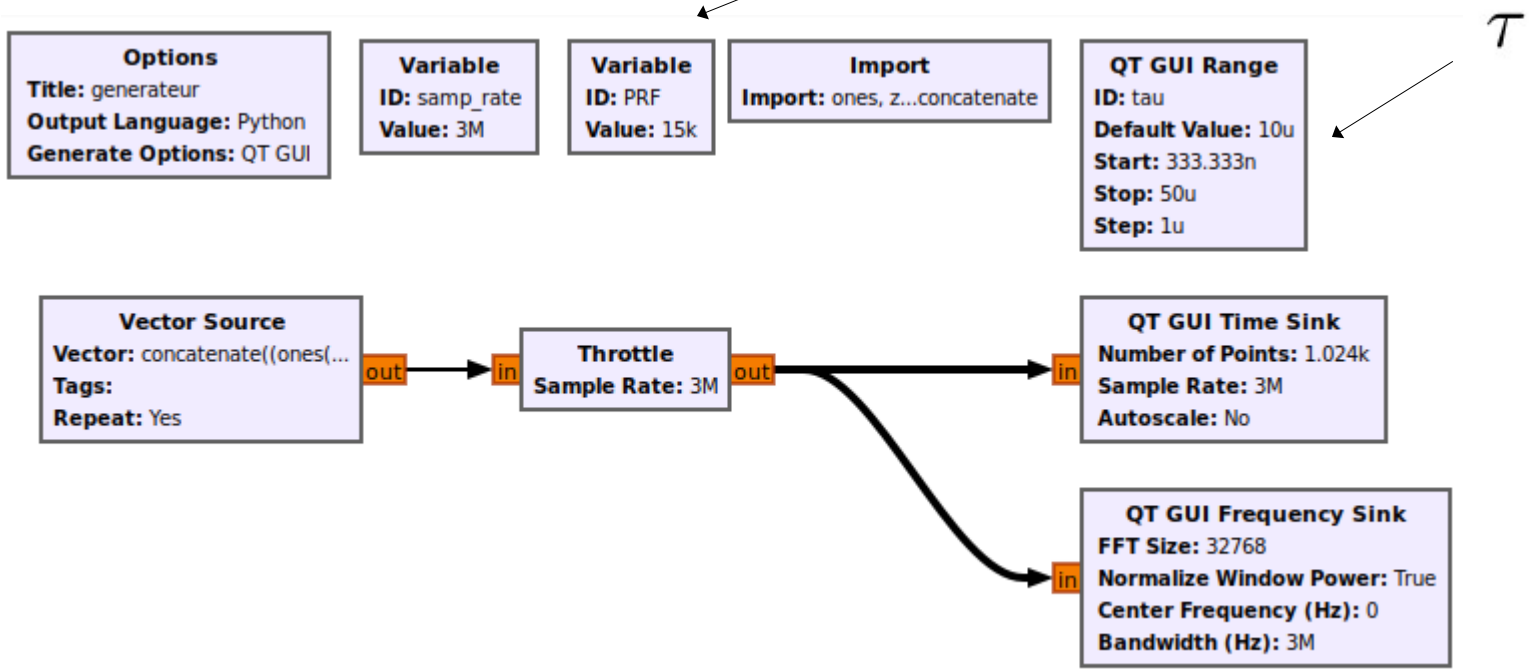
Présentation

Manipulation proposée



Forme temporelle : signal en bande de base

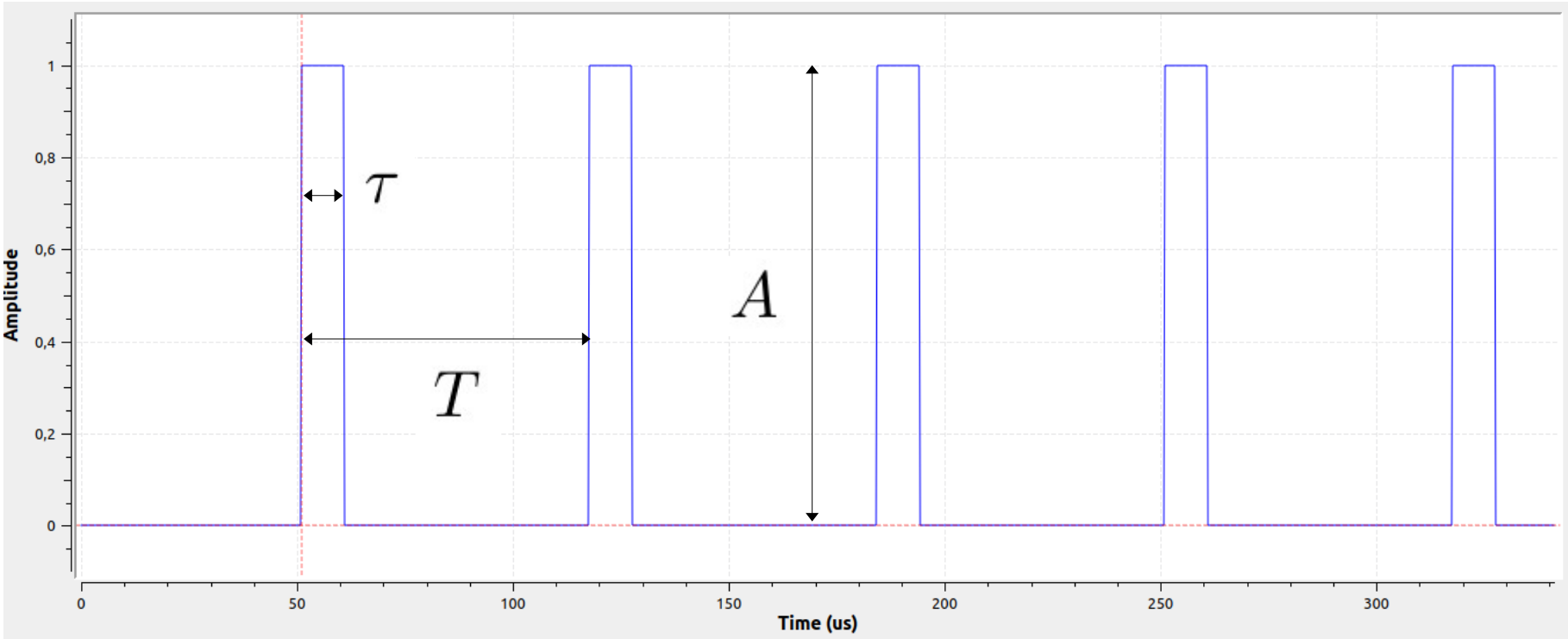
$$PRF = \frac{1}{T}$$



τ : **Largeur d'impulsion (Pulse width)**

T : **Période de répétition (Pulse delay)**

Forme temporelle : signal en bande de base



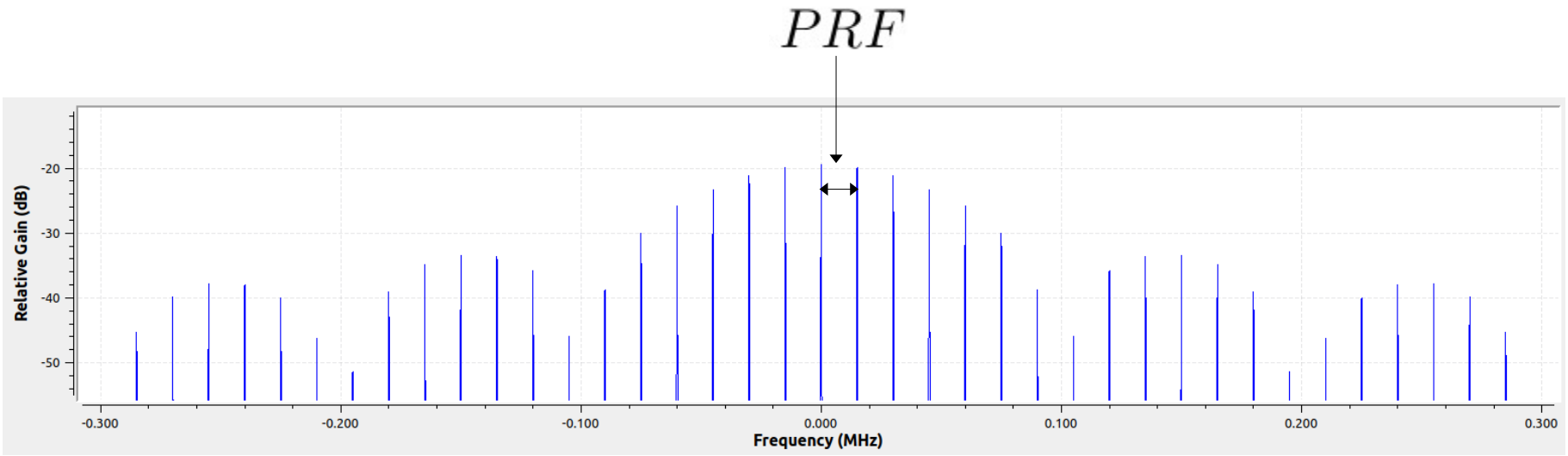
τ : **Largeur d'impulsion (Pulse width)**

T : **Période de répétition (Pulse delay)**

A : **Amplitude**

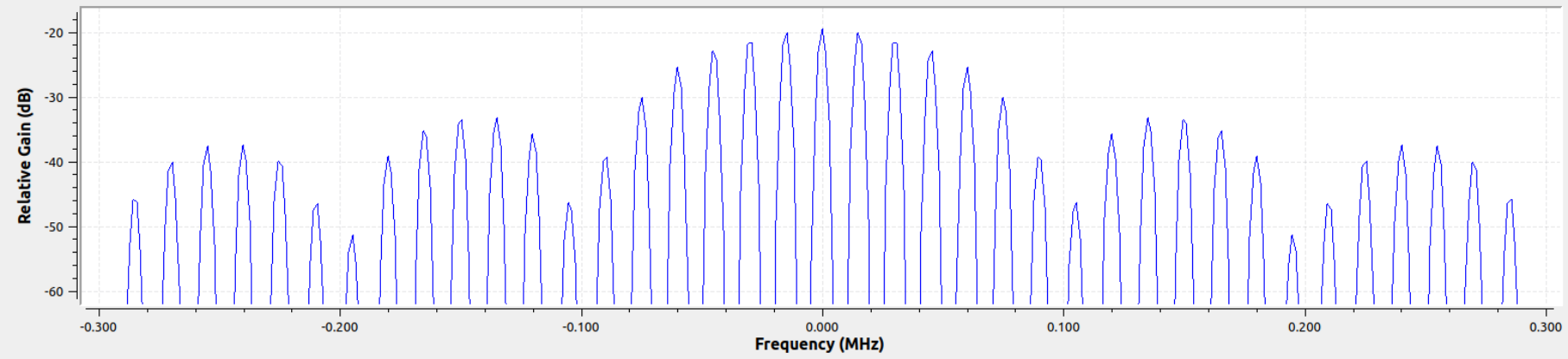
$$PRF = \frac{1}{T}$$

Forme fréquentielle : signal en bande de base

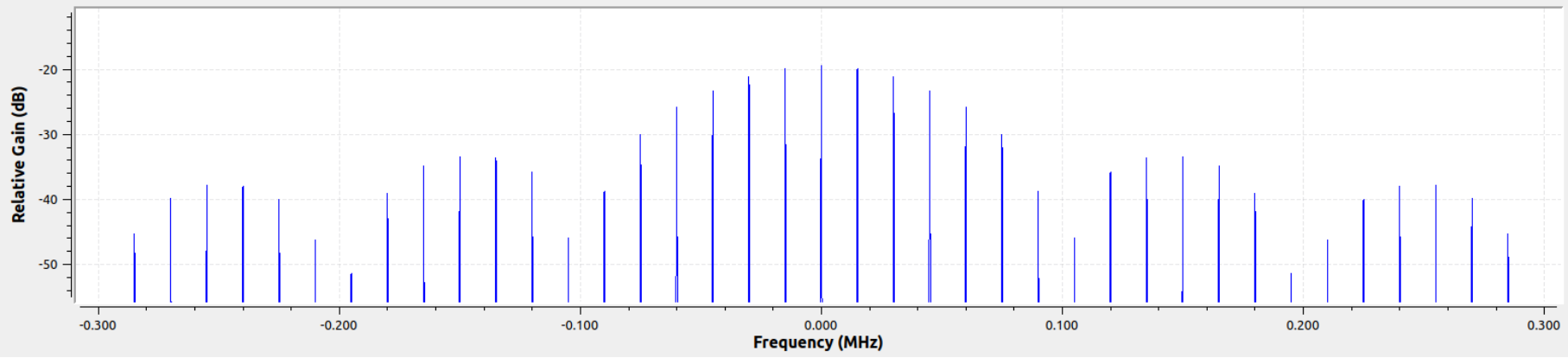


Forme fréquentielle : signal en bande de base

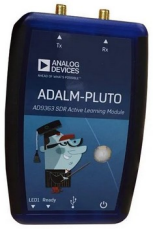
Durée d'observation : 0.682 ms

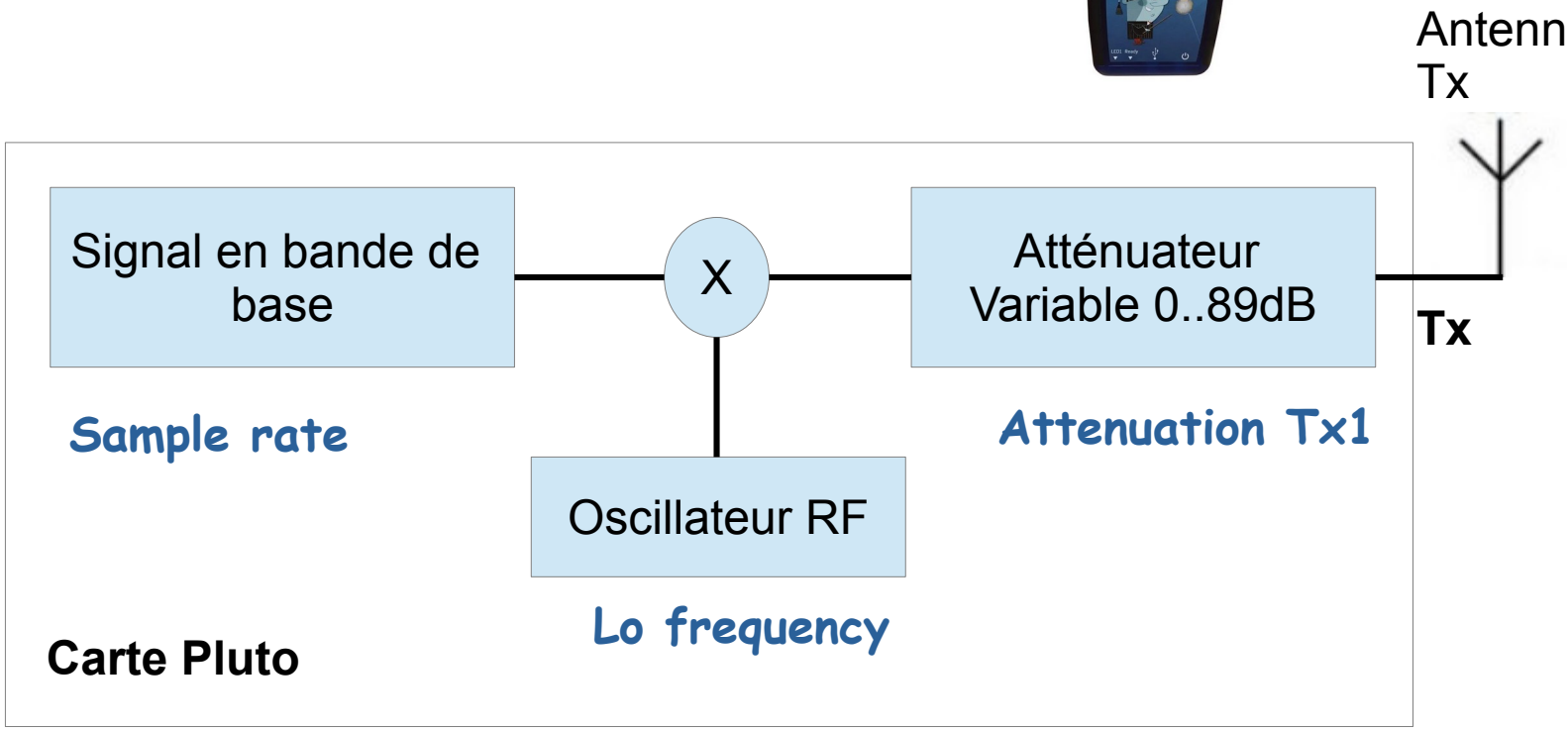


Durée d'observation : 10.9 ms



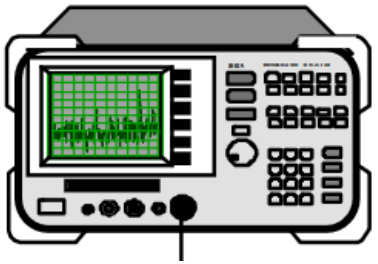
Emetteur : Carte ADALM PLUTO





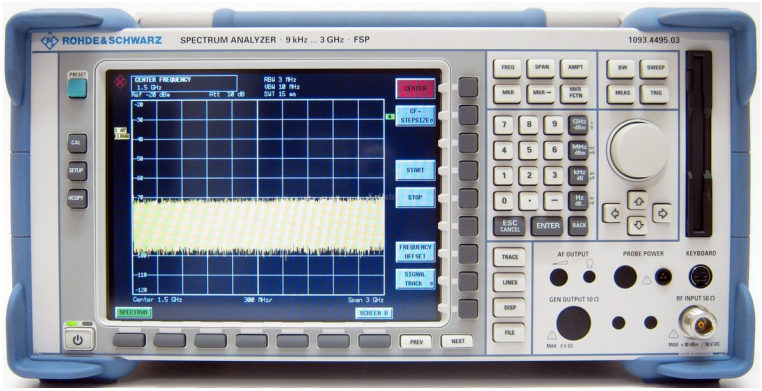
- **Lo frequency** : Fréquence RF du signal radar (ex 2.4GHz)
- **Attenuation Tx1 (dB)** : conditionne la puissance de transmission du radar (max 10dBm)
- **Sample rate** : lié à la bande passante du signal transmis (Sample rate > $\frac{2}{\tau}$)

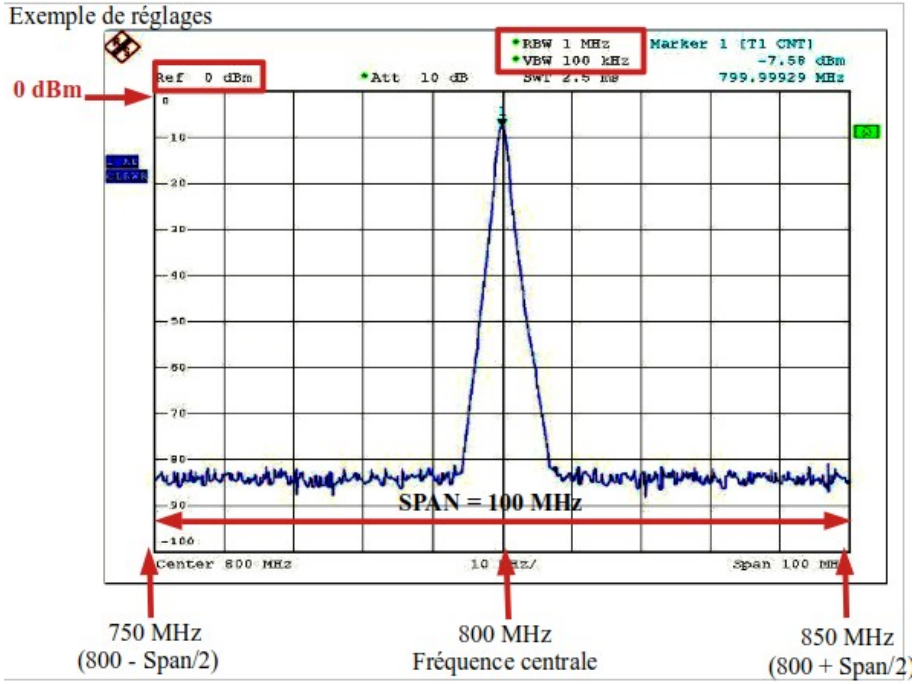
Récepteur : Analyseur de spectre



Qu'est-ce qu'un analyseur de spectre ?

- Permet de mesurer la puissance d'un signal (son amplitude) sur une plage de fréquence donnée.
- En utilisant la fonction « zéro span », il permet d'afficher la variation de la puissance en fonction du temps (pour une fréquence fixée).

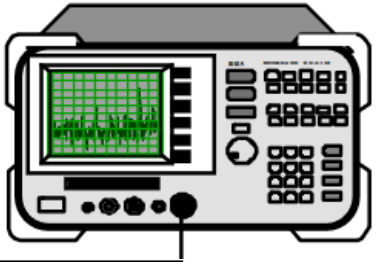




Aspect expérimental

Manipulation simplifiée :

Analyseur de spectre
FPL1003 5KHz-3GHz

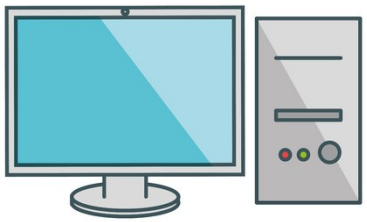


Atténuateur 20 dB



SDR
Pluto

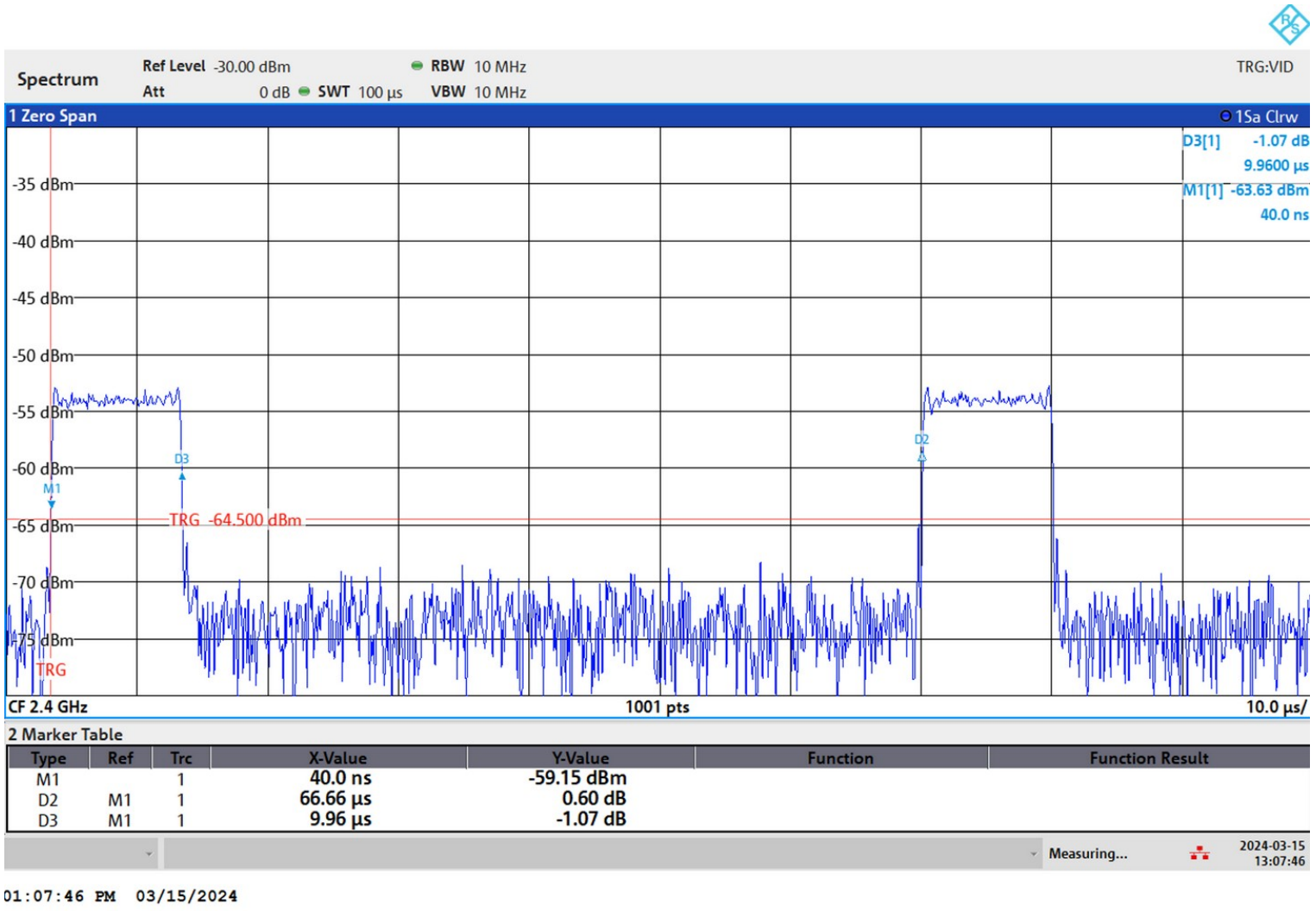
Cordon SMA



Pulse width : variable
PRF = 15 Khz → delay = 67us

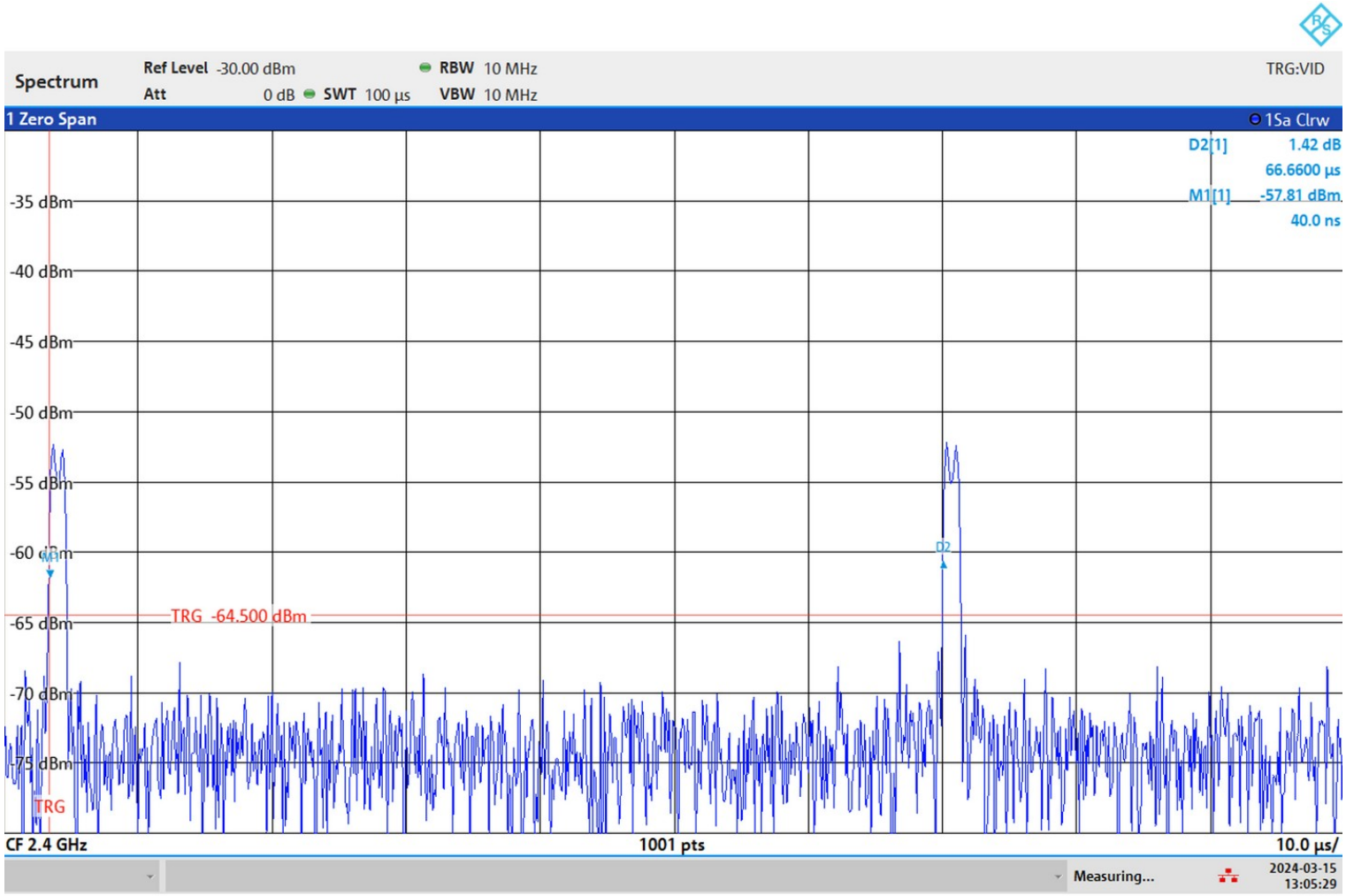
Domaine temporel : zéro span

Impulsion 10 us



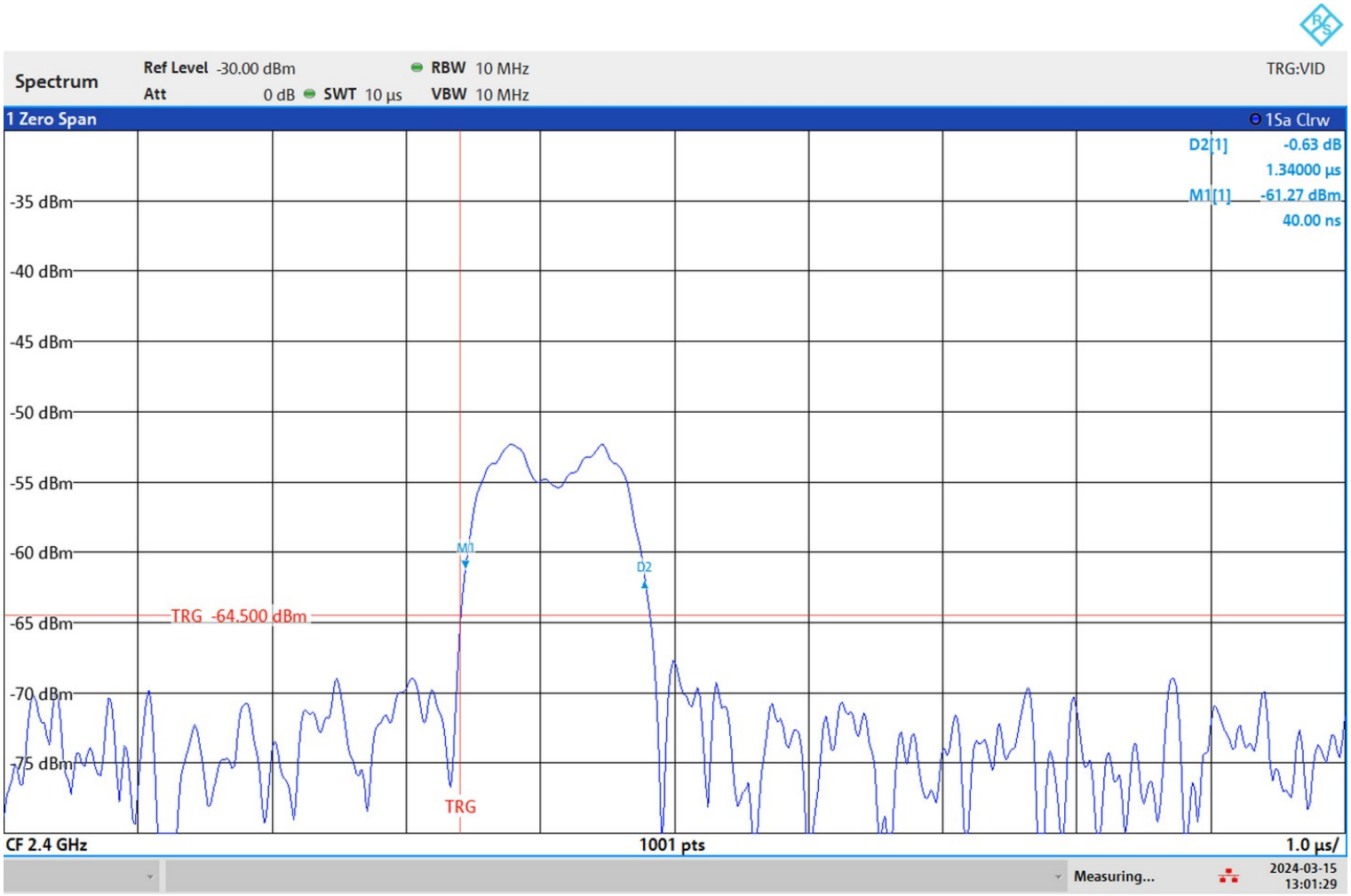
Domaine temporel : zéro span

Impulsion 1.3 us



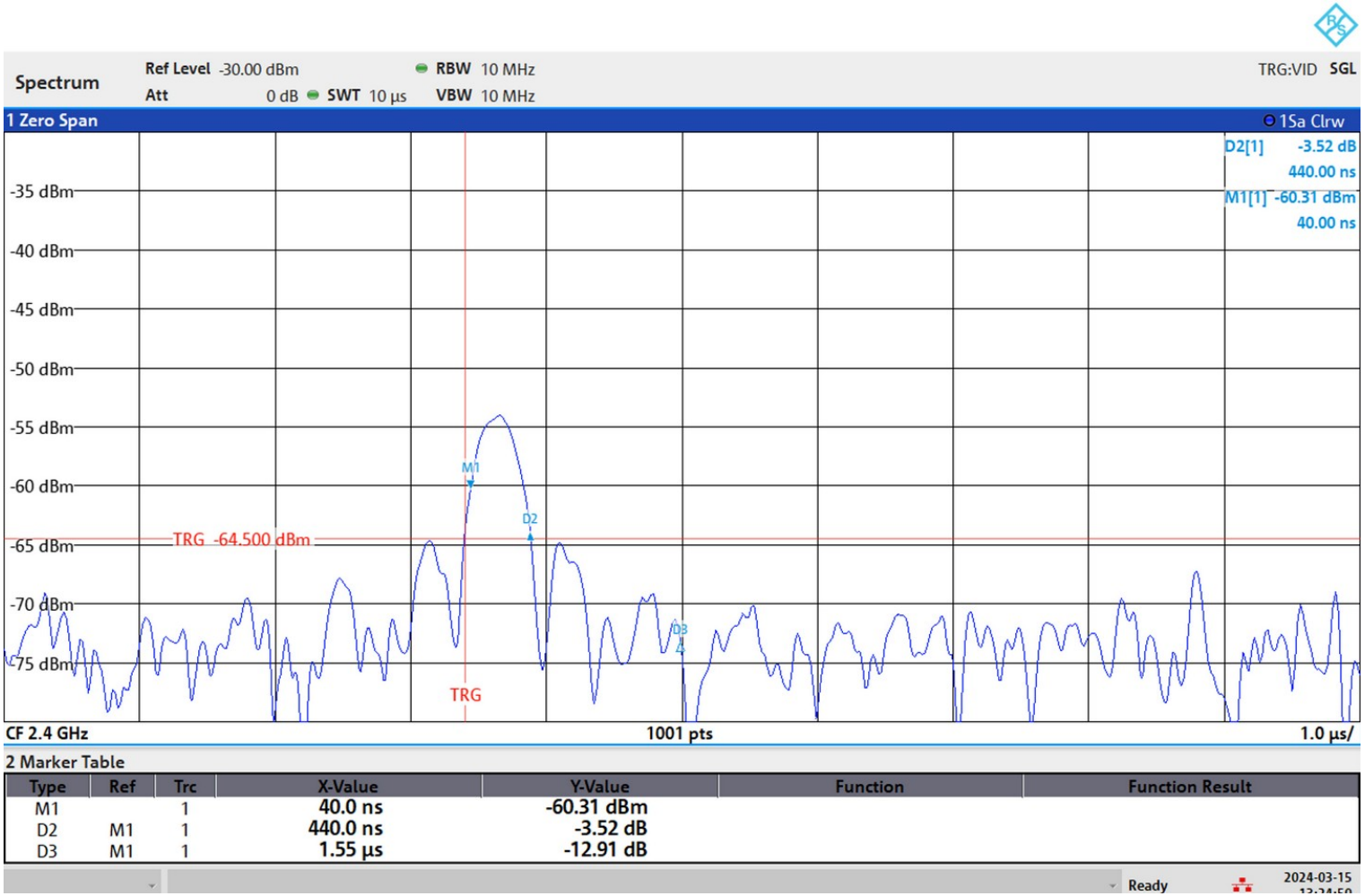
Domaine temporel : zéro span

Impulsion 1.3 us



Domaine temporel : zéro span

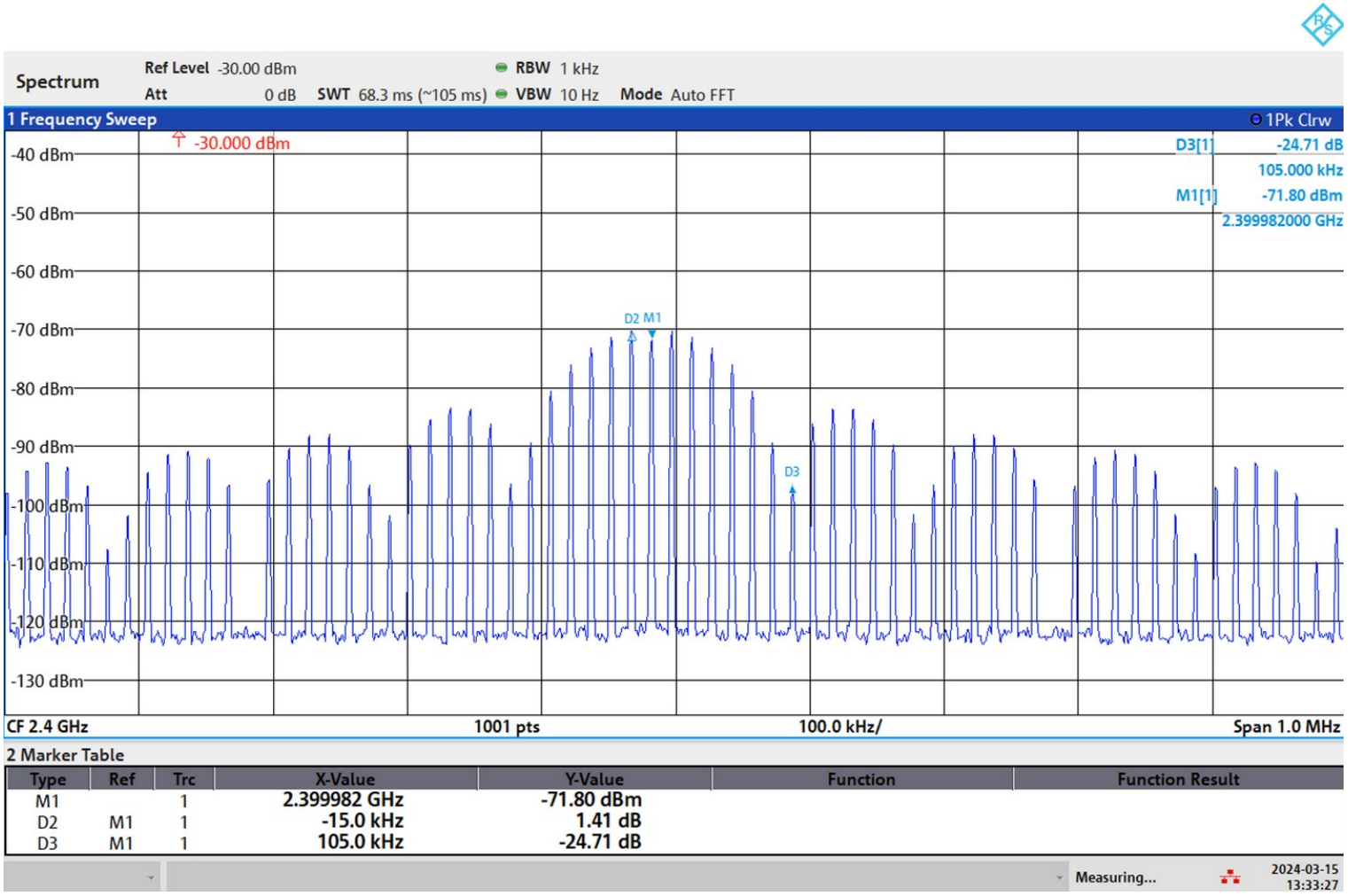
Impulsion 0.4 us ?



Aspect expérimental

Domaine fréquentiel :

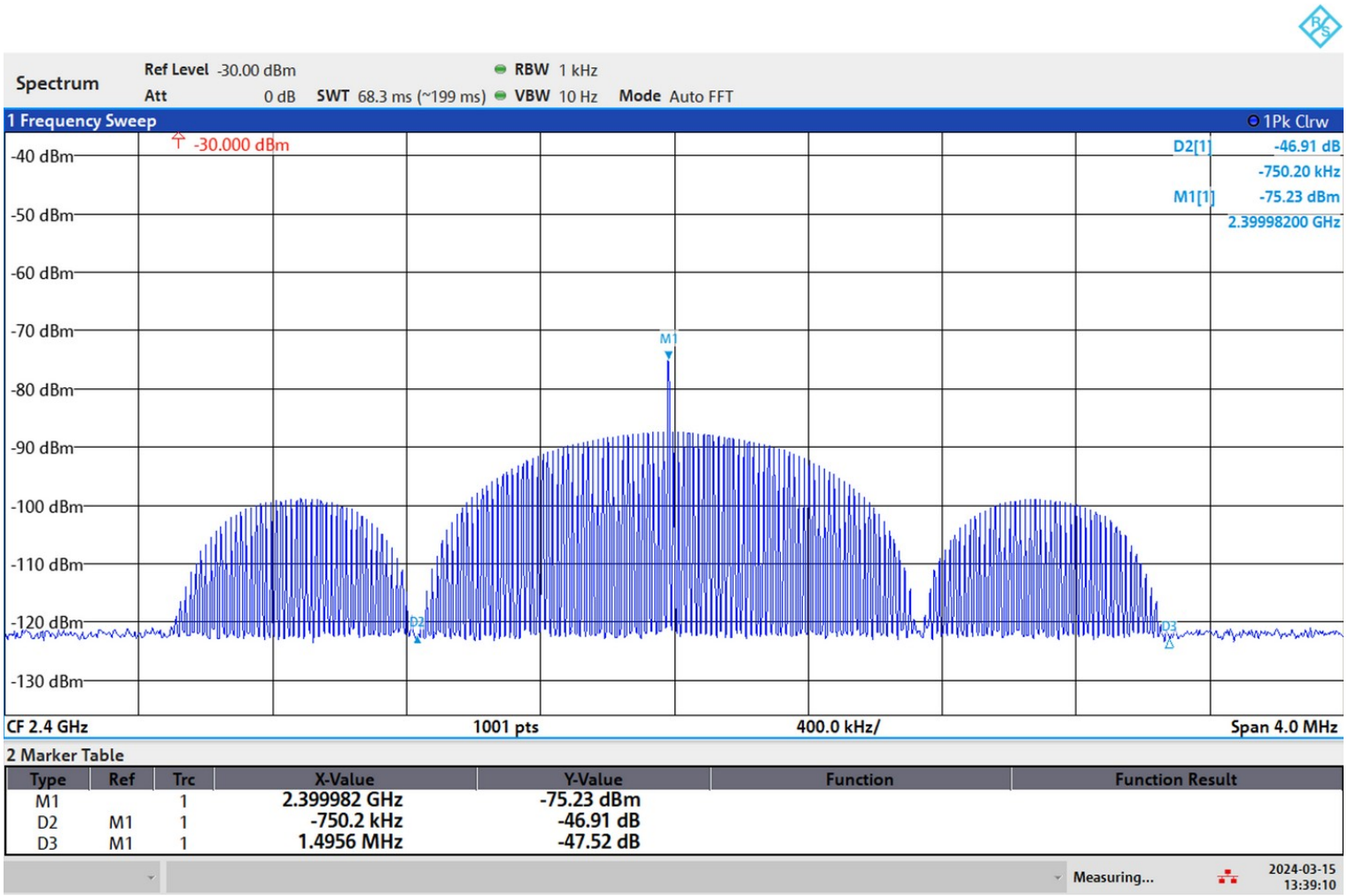
Impulsion 10 us



Aspect expérimental

Domaine fréquentiel :

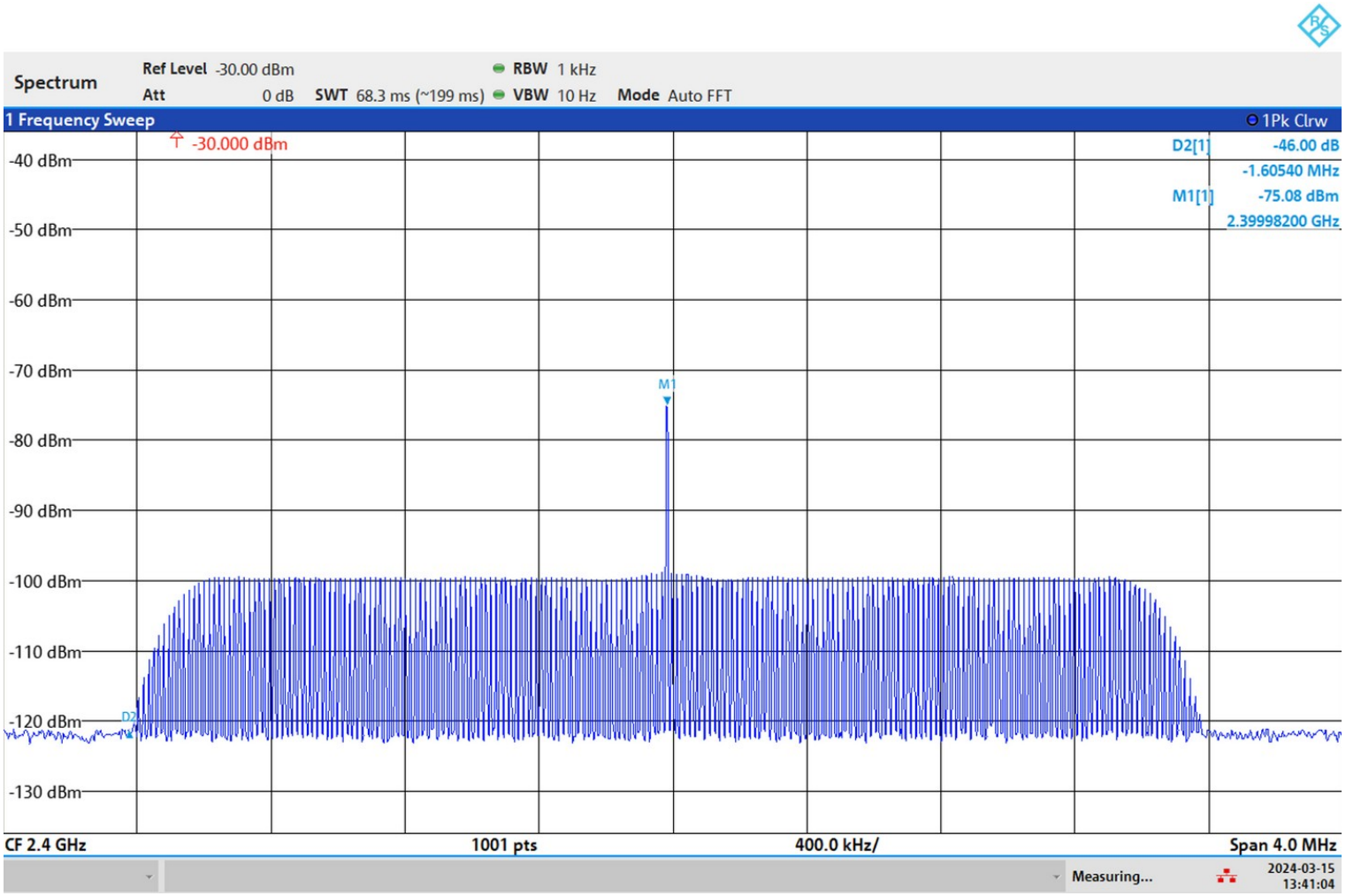
Impulsion 1.3 us



Aspect expérimental

Domaine fréquentiel :

Impulsion 0.4 us



Aspect expérimental

Domaine fréquentiel :

Carte pluto branchée mais gnuradio companion stoppé

